Measuring Power Output for Characterizing Unknown Specifications of Heterodyne Receivers in Radio Astronomy

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1. BACKGROUND

- Radio astronomy is at the forefront of current astronomical research. Many telescope arrays are bult around the world to produce high-resolution images of astronomical objects.
- Most arrays operate in the range of 80 to 270 GHz. Our motivation for this project is to find a method to reduce these large signals to manageable frequencies to generate images.



CARMA Array – Caltech, Credit: Nick Sower

2. INTRODUCTION

- Radio telescopes collect and process light. This is done via numerous electronic components that manipulate and refine light after collection.
- Most of these components are for microwaves in the range of 1-2 GHz.
- We will be looking at the heterodyne receiver, responsible for mixing (heterodyning) high, unmanageable **radio frequencies (RF)**, with a **local oscillator (LO)** signal to produce a lower, fixed **intermediate frequency (IF)** that is easy to manage.
- Our goal is to devise a method to create a frequency that is as strong as possible, as it is difficult to keep large signals accurate. However, more power means it is simpler if the signal has a reference to "lock" to.

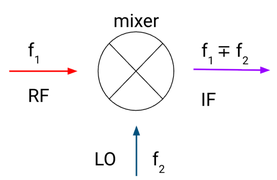


Fig 1. A diagram of the process of heterodyning.

3. SETUP

- We worked with 90/230 GHz receiver systems acquired from the CARMA array. These systems are currently in the works of being reactivated for the Arizona array.
- Figure 2 describes our setup for this project. It consists of a **Gunn diode generator** on the far left with two knobs used to tune the frequency and power of the incoming signal. A series of mixers, couplers, attenuators, and amplifiers are used to direct the signal into a power meter, which we measured.
- Before the setup, a multistep signal generating process detailed in the following section and in Figure 3 is used to generate a large signal, which is difficult to achieve.

4. METHODS

1. Figure 3 describe our step process. Through a LO generator, a signal is generated in the range of 1 GHz. We specifically look at **harmonics** (integer multiples) of that signal that corresponds to a range of 8-12.5 GHz, used to phase lock into a **yttrium-iron-garnet (YIG) reference generator**. We need to know the exact harmonic this signal corresponds to the YIG, to derive consistent relationships between harmonics and **digital analog converter (DAC) voltage** being applied.
2. The signal is then locked to the 8th harmonic of the YIG. We used both the 8th and 9th harmonic of the YIG's frequency to phase lock the signal into the Gunn. This IF is predicted, and to confirm if this frequency corresponds to the Gunn, we need to ensure that this frequency corresponds to the greatest power output. Figure 4 confirms if our signals for each generator are phase locked.
3. After obtaining a linear relationship of frequencies corresponding to the YIG in terms of DAC applied, a similar process is applied to determine the harmonics of the Gunn oscillator. Tuning the Gunn to a phase-locked harmonic is done by changing the **frequency knob**, which changes the resonant frequency of the signal within the Gunn. Once resonance is achieved, we optimized power with the **backshort knob**.
   - The backshort is a piece of metal located inside the Gunn that changes position as the knob is turned, causing the signal to switch between odd numbers of quarter wavelengths. A power meter is then used at the end of the setup to measure the power of the final signal.
4. Once we determine the maximum power for certain harmonics in the YIG and Gunn, by deriving a relationship between DAC and frequency we can determine the right frequency the Gunn outputs.

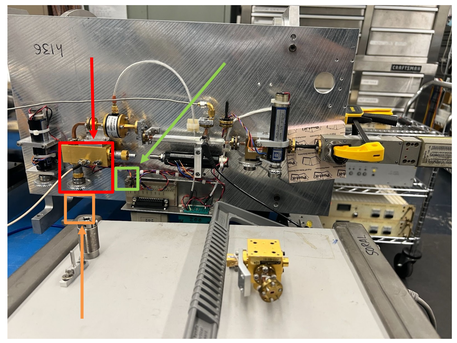


Fig 2. Photo of our receiver system, consisting of a Gunn diode generator towards the left of the image, indicated by the bright red arrow. Two important components, the frequency and backshort knob, are used to tune the Gunn in our setup. They are indicated by the green and orange arrows, respectively. The Gunn is then connected by a series of circuit components to direct the signal. Lastly, a power meter at the end to the right of the image is attached using clamps to measure power output (in mW).

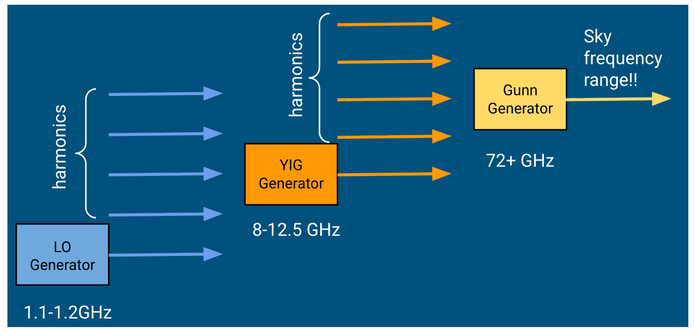


Fig 3. A diagram of the multistep signal process we used for our project. Harmonics of the main signal is generated by nonlinearities in the system is then phase locked corresponding to frequencies in the range for each generator. For a certain harmonic that is within the right frequency for each generator, it is then accounting for a frequency offset. This is applied to both the YIG and Gunn to create higher frequency signals from a low frequency one. This process is crucial into giving us an idea of predicting what frequencies the Gunn operates for this specific receiver.

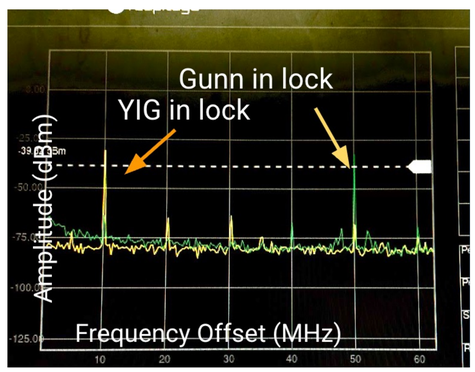


Fig 4. A GUI that shows the signals for each generator in phase lock. The frequency offset for both generators is intentional in order to reduce interference. They are defined to be 10 MHz and 50 MHz for the YIG and Gunn respectively.

5. RESULTS

- Following our process, we first characterized the YIG. Using the equation $f_L = N f_{YIG} \pm 10\,MHz$, we determined harmonics of frequencies from 1100-1260 MHz and measured DAC for each one to create a linear plot of DAC vs. YIG frequency below.

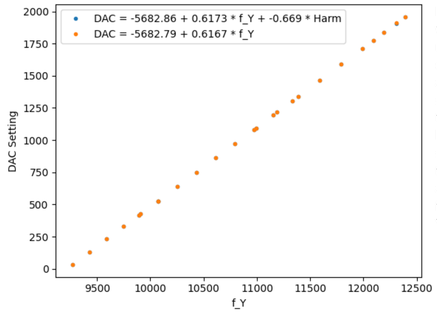


Fig 5. Linear plot of DAC vs. YIG frequency. We used matrix methods to devise an equation that allows us to pick a DAC setting that corresponds to a phase-locked YIG harmonic signal. For this specific case, we used the 8th harmonic that allows us to satisfy frequency ranges of the YIG.

- Similarly, for the Gunn, we used $f_{YIG} = N f_{GUNN} \pm 50\,MHz$ to determine possible Gunn frequencies. Using the above equation and the derived relationship in Figure 6, we determined the GUNN oscillator to operate in the range of 72-90 GHz.

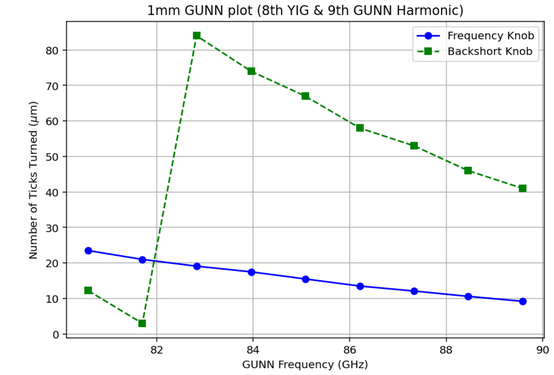


Fig 6. Plot of tick marks (measured in micrometers) of how much the frequency and backshort knob was turned to achieve a phase lock, versus Gunn frequencies predicted by our calculations. We found that using the 8th & 9th harmonics of the GUNN gave us frequencies in the range of 72-90 GHz combined. The plot above is for the 9th harmonic, covering the upper range of possible Gunn frequencies.

6. CONCLUSIONS

- We have successfully determined the frequencies the GUNN oscillator we used, which was previously unknown.
- Moving forward, our measurements and data collection methods will be used for other ongoing receiver testing efforts.
- We will focus on tuning the mixer to ensure little noise is generated now that we know the most optimal conditions, important for accurate astronomical observations.

REFERENCES

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Plambeck, Dick, et al. *CARMA Antenna LO Plan Antenna LO System Requirements*. 2 Mar. 2002.

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